

Individual Assignment 5

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Set up

Packages

```
#general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(palettter)
library(ggthemes)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

copied from class

```
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include sites in ncos_sites
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>
```

```

# create new columns that show median of each variable at each date_site
# because there are multiple samples at the same site at different depths
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

# ungroup
ungroup() |>

# filter out the salinity outlier
filter(date_site != "2022-08-12_NVBR")

```

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

# converted taxon name values to snake case (lowercase)
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))

```

```

# creating new clean aquatic inverts object from aquatic inverts
aq_ins_clean <- aq_ins |>

# cleaning column names
clean_names() |>

# filtering join: only include quarterly NCOS surveys from ncos_sites
semi_join(ncos_sites, by = "site") |>

# renaming column to just date
rename(date = date_on_vial) |>

# select columns of interest
select(date, sample_type, site,

       # 9 most abundant taxon
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,

```

```

    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>

# filter to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250")) |>

pivot_longer(
  # cols = columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>

# group by the date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

# ungroup
ungroup() |>

# create a new column called `date_site` to join with water quality
# don't remove the original columns
unite("date_site", date, site, remove = FALSE) |>

# join with water quality clean
# wqc has `date_site` column as well
left_join(water_quality_clean, by = "date_site") |>

# join with taxonomic information from taxon list
# verbatim_name column in taxon_list_clean matches with taxon column in
  ↪ aq_ins_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",

```

```

site == "NEC" ~ "East Channel",
site == "NMC" ~ "Main Channel",
site == "NPB" ~ "South of Phelps Bridge",
site == "NPB1" ~ "North of Phelps Bridge",
site == "NPB2" ~ "Phelps Road",
site == "NWP" ~ "West Pond",
site == "NDC" ~ "Devereux Creek"
)) |>

# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
↪   = TRUE),
       site_full = fct_rev(site_full))

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: average abundance of ostracod per liter
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```

# creating new object from aquatic insects
ostracod <- aq_ins_clean |>

# filter to only include ostracod
filter(taxon == "ostracod") |>

# calculate the average of ostracod abundance per lit
mutate(ostracod_log_avg_per_lit = log(avg_per_lit + 1))

# display 10 random rows
slice_sample(ostracod, n = 10)

```

```
# A tibble: 10 x 24
  date_site date site taxon avg_per_lit med_ph med_sal med_do
  <chr> <dtm> <chr> <chr> <dbl> <dbl> <dbl> <dbl>
1 2022-08-12~ 2022-08-12 00:00:00 NVBR ostr~ 0 NA NA NA
2 2021-11-12~ 2021-11-12 00:00:00 NPB ostr~ 79.2 NA NA NA
3 2022-08-08~ 2022-08-08 00:00:00 NPB2 ostr~ 2 7.68 2.36 2.43
4 2022-05-12~ 2022-05-12 00:00:00 NPB ostr~ 1.07 7.97 11.4 8.17
5 2022-05-12~ 2022-05-12 00:00:00 NPB2 ostr~ 1198 NA NA NA
6 2022-08-12~ 2022-08-12 00:00:00 NEC ostr~ 0 7.37 NA NA
7 2023-06-06~ 2023-06-06 00:00:00 NPB1 ostr~ 0.533 7.48 NA 2.38
8 2020-02-21~ 2020-02-21 00:00:00 NVBR ostr~ 4.8 7.34 4.1 6.1
9 2022-05-06~ 2022-05-06 00:00:00 NMC ostr~ 0 9.96 1.45 16.5
10 2023-02-28~ 2023-02-28 00:00:00 NPB ostr~ 2.53 NA NA NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, ostracod_log_avg_per_lit <dbl>
```

c. Code your figure

```
# base layer: create an object that stores the plot
ostracod_plot <- ggplot(data = ostracod,
  # x = site
  # y = log + 1 transformed average abundance/L
  mapping = aes(x = site_full,
    y = ostracod_log_avg_per_lit)) +

# first layer: geometry jitter points to represent each sample
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +

# points to represent median abundance
stat_summary(geom = "point",
  fun = median,
  color = "green",
  size = 2) +

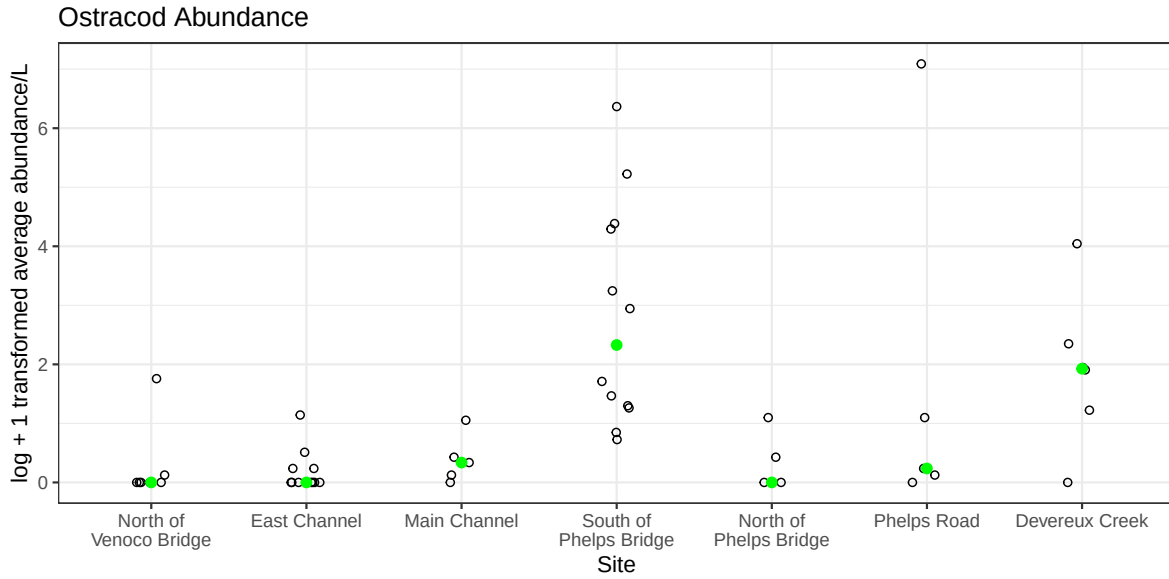
# wrapping x axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +

# labeling axes and title
labs(title = "Ostracod Abundance",
```

```
x = "Site",
y = "log + 1 transformed average abundance/L") +

# changed theme
theme_bw()

# display plot
ostracod_plot
```



- d. Create a multipanel figure
- e. Interpret your figure

2. Visualizing total abundance as a function of salinity

- a. Plan your figure
- b. Wrangle your data
- c. Code your figure
- d. Interpret your figure